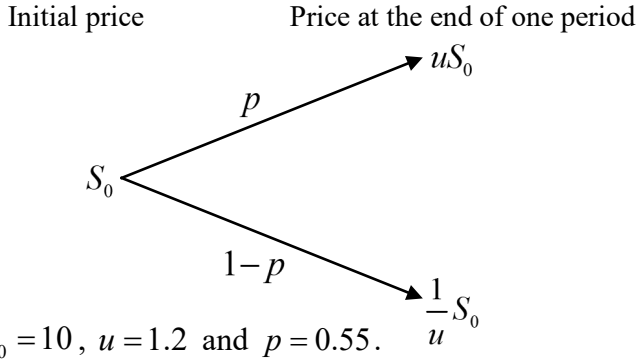
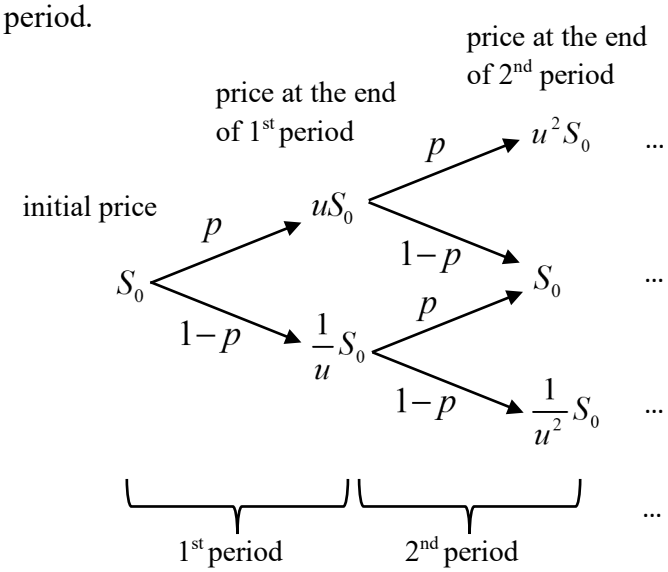
RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision (Zeta)
Topic(s): (C) Statistics

1 **(a)** In a one-period binomial model, the stock price initially at S_0, either rises to uS_0 with a probability p or drops to $\frac{1}{u}S_0$ with a probability $(1-p)$ at the end of one period, where $u>1$.



Suppose that $S_0=10$, $u=1.2$ and $p=0.55$.
Show that the expected value of the stock price S at the end of the period is 10.35 and find the variance of the stock price. [3]

(b) The above model can be extended to a n-period binomial model. For each period, the stock price S either rises to uS with a probability p or drops to $\frac{1}{u}S$ with a probability $1-p$ at the end of this period.



(i) In the first quarter, the initial price of the stock at the start of the first period is S_0. Suppose the stock price rises 4 times and drops twice over 6 periods. Find the stock price in terms of S_0 and u at the end of the 6 periods. [1]

(ii) Show that the probability for part **(b)(i)** to happen is $15p^4(1-p)^2$. [1]

(iii) In the second quarter, the initial price of the stock at the start of the first period is S_1. Find the probability that the stock price is higher than S_1 at the end of 6 periods in terms of p. [2]

(c) A *contract* that gives investors the right to buy a stock at a specified *price* at the end of a specified period is termed "call option". The specified price is called "strike price" and the amount needed to buy this contract is called "premium".

Each unit of a stock's call option has a return value V at the end of the period. Its value is given by

$$V = \begin{cases} S - S_{\text{strike}} & \text{if } S > S_{\text{strike}}, \\ 0 & \text{if } S \le S_{\text{strike}}, \end{cases}$$

where S_{strike} is the strike price and S is the stock price at the end of the period.

Stock K, currently priced at \$10, is modelled by the n-period binomial model in part **(b)** with $n = 6$, $u = 1.2$ and $p = 0.55$. A call option of stock K has a strike price of \$10.

(i) Fill in the missing numbers in the following probability distribution table of V for this call option, giving the values to a suitable degree of accuracy. [2]

Number of times for the stock price to rise	3 or below	4	5	6
v	0		10.736	19.85984
$\mathrm{P}(V = v)$			0.13589	0.02768

The call option currently has a premium of \$2.50 per unit. Peter plans to invest \$10 000 on purchasing 4000 units of the call option and sell them at the end of the 6 periods.

(ii) Find the expected return value of the investment, giving your answer to the nearest dollar. [2]

(iii) Comment on whether Peter should proceed with this investment plan. [1]

2 On average, 7.8% of markers produced by a factory are faulty. Each day, the factory's quality manager picks a random sample of n markers and inspects the number of faulty markers found in the sample. The number of faulty markers found in the sample is the random variable X and whether the markers are faulty are independent of one another.

(i) Explain what is meant by a random sample in this context. [1]

(ii) Given that the probability that more than $(n-4)$ markers in the random sample are found to be non-faulty is less than 0.3, find the least possible value of n. [2]

Assume for the remainder of this question that $n = 10$.
The samples in 50 randomly chosen days are collected and the number of faulty markers in each sample is recorded.

(iii) Find the probability that exactly 35 of the samples each contains at least one faulty marker, given that none of the 50 samples contain more than 2 faulty markers each. [4]

3 In this question you should state clearly all the distributions that you use, together with the values of the appropriate parameters.

In a factory, light oil and heavy oil are stored in barrels of different sizes. It is given that the volumes of oil in the barrels can be modelled using normal distributions with means and standard deviations as shown in the table.

	Mean volume (in litres)	Standard deviation (in litres)
Light oil	110	2.5
Heavy oil	145	3.5

(i) Find the probability that the volume of light oil in a randomly chosen barrel is between 104 litres and 116 litres. [1]

(ii) Sketch the distribution for the volume of light oil, indicating clearly the probability found in part **(i)**. [2]

(iii) A random sample of seven barrels of heavy oil is chosen. Find the probability that exactly four barrels of heavy oil have volume between 142 and 150 litres each and exactly one barrel of heavy oil has volume more than 150 litres. [3]

(iv) A random sample of n barrels of heavy oil is chosen and it is given that the probability that the mean volume of these n barrels of heavy oil exceeding k litres is at least 0.3. Find an inequality involving k and n. [3]

A lorry with a maximum laden mass of 6800 kilograms is used to transport 25 barrels of light oil and 30 barrels of heavy oil. It is given that the densities of light oil and heavy oil are 0.83 kilograms per litre and 0.94 kilograms per litre respectively, and the empty barrels for light oil weigh 5 kilograms each and the empty barrels for heavy oil weigh 8 kilograms each.

(v) Find the probability that the load of the lorry exceeds its maximum laden mass. [4]

[Density is defined as $\frac{\text{Mass}}{\text{Volume}}$.]

(vi) State an assumption needed for your calculations in part **(v)**. [1]

4 A biased 4-sided die gives the scores 1, 2, 3, and 4 with the probabilities shown in the table, where *a* and *b* are constants. The random variable S denotes the score on the die.

Score (s)	1	2	3	4
$\mathrm{P}(S=s)$	a	a	a	b

It is given that the mean of the score is 2.56.

(a) Show that $b=\frac{7}{25}$. [2]

(b) S_1 and S_2 are two independent observations of *S*.

(i) Find the probability that the sum of the two independent observations of S is at least 6, given that at least one of the observed scores is 3, [3]

Let $Y=\left|S_1-S_2\right|$. Without finding the probability distribution table of Y,

(ii) find $\mathrm{Var}\left(2S-\mathrm{E}(Y)\right)$. [3]

(iii) state the value of $\mathrm{E}\left(Y-\mathrm{E}(Y)\right)$. [1]

5 A company produces light bulbs with a brightness rating of 470 lumens. It is known that the brightness of these light bulbs is normally distributed with a standard deviation of 10 lumens. A production manager wishes to test whether the mean brightness is less than 470 lumens. He collects a random sample of 40 light bulbs and finds that their mean brightness is 466.8 lumens.

(i) Explain what is meant by the term "random sample" in the context of this question. [1]

(ii) Carry out a test at the 5% level of significance for the production manager, clearly stating the *p*-value of this test. Explain what *p*-value means in the context of this question. [5]

(iii) Suppose that the production manager had tested if the mean brightness differed from 470 lumens at the 5% level of significance. Without performing another hypothesis test, determine whether your conclusion in **part (ii)** would be affected. [2]

The company decides to produce a new line of 'lowlight' bulbs with a brightness rating of 350 lumens. A random sample of 50 'lowlight' bulbs is taken and the brightness level, *y* lumens, are summarised as follows.

$$n=50 \qquad \sum(y-350)=-125 \qquad \sum(y-347.5)^2=4704$$

(iv) Calculate unbiased estimates of the population mean and variance for the brightness of the 'lowlight' bulbs. [2]

(v) Given instead that the standard deviation of the brightness of the 'lowlight' bulbs is known to be σ, the company found insufficient evidence from this sample to conclude that the mean brightness was less than 350 lumens at the 1% level of significance. Determine the range of values of σ. [2]

6 In this question you should, where applicable, state clearly all the distributions that you use, together with the values of the appropriate parameters.

A supermarket produces its own packs of white sesame seeds and fennel seeds for sale under its house brand. The masses (in grams) of a randomly chosen pack of white sesame seeds and that of a randomly chosen pack of fennel seeds, denoted by W and F respectively, have independent normal distributions. The means and standard deviations of these masses are shown in the following table.

	Mean (g)	Standard deviation (g)
W	250	σ
F	300	2.5

It is given that 5% of the packs of white sesame seeds have a mass of less than 245 grams.

(a) Show that the value of σ is 3.0398, correct to 5 significant figures. [2]

(b) Find the probability that the total mass of six randomly chosen packs of white sesame seeds is less than five times the mass of a randomly chosen pack of fennel seeds by not more than 20 grams. [3]

(c) n packs of white sesame seeds and n packs of fennel seeds are chosen at random. Find the smallest value of n such that the probability that the mean mass of these $2n$ packs being at least 278 grams is less than 0.015. [4]

7 Researchers at University S claimed that *Fungus A*, a strain of fungi that is typically found in soil and plants, was able to break down polypropylene, a plastic that is hard to recycle, that had been pre-treated with either UV light or heat by at least 27 grams per 100 grams of polypropylene over 90 days.

Researchers at University M believe that the researchers at University S overstated their claim and decided to perform a hypothesis test on a random sample of 120 batches of polypropylene, each of mass 100 grams. The researchers recorded the amount x grams of polypropylene for each batch that *Fungus A* was able to break down over a 90-day period. The results are summarised as follows:

$$\sum(x-27)=-81,\ \sum(x-27)^2=2070.8$$

(a) Calculate the unbiased estimates of the population mean and variance. [3]

(b) Test, at 5% level of significance, the claim that the population mean mass of polypropylene per 100 grams that was broken down is at least 27 grams. You should state your hypotheses and define the symbols you use. [4]

(c) State, in the context of the question, what you understand by the expression 'at 5% level of significance' found in part **(b)**. [1]

(d) Explain if it is necessary for University M to assume a normal distribution for the mass of polypropylene per 100 grams that *Fungus A* was able to break down for this test to be valid. [1]

University S responded by experimenting on 50 batches of polypropylene, each of mass 100 grams, and recorded the amount y grams of polypropylene for each batch that *Fungus A* was able to break down over a 90-day period. The null and alternative hypotheses remain the same as the ones stated by University M in part **(b)**.

Based on the research findings from University S, the population standard deviation of the mass of polypropylene per 100 grams that *Fungus A* was able to break down is known to be 4 grams.

(e) Find the set of values of sample mean mass, $\overline{y}$ grams, of polypropylene per 100 grams from University S such that the null hypothesis is not rejected in favour of the alternative hypothesis at 5% level of significance. [3]